Fiscal Year:	FY 2024	**Task Last Updated:**	FY 03/27/2024
PI Name:	Moudy, Sarah Ph.D.		
Project Title:	Development of Sensorimotor Fitness for Duty Assessments Using Ground Analogs		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
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Space Biology Special Category:	None		
PI Email:	sarah.c.moudy@nasa.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	
Organization Name:	KBR Wyle/NASA Johnson Space Center		
PI Address 1:	2400 NASA Parkway		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
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Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	This investigation is a continuation of a Directed Research project with the same name under PI Marissa Rosenberg. Dr. Rosenberg has left NASA. The new PI is Sarah C. Moudy, Ph.D. with KBR/Aegis Aerospace. Millard Reschke, Ph.D. has retired from NASA so he no longer serves as Co-Investigator.		
COI Name (Institution):	Wood, Scott Ph.D. (NASA Johnson Space Center) Peters, Brian Ph.D. (NASA Johnson Space Center) Clark, Torin Ph.D. (University of Colorado, Boulder) Schubert, Michael (Johns Hopkins University)		
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Task Description:	The first aim will develop a Sensorimotor Disorientation Analog (SDA) that can provide different levels of acute disorientation through combined vestibular, visual, and proprioceptive disruptions. The SDA will be used to increase the range of performance, simulating the moderate-to-severe performance decrements observed postflight. The levels of SDA will be titrated and validated by 1) using feedback from experienced astronauts, and 2) comparison to gold standard measures that have a wealth of spaceflight data at different time points during recovery. Astronaut postflight experience will provide a more realistic starting point for validation of the SDA. Referencing existing postflight data from the gold standard measures will also help us characterize how each magnitude of SDA compares to recovery from long-term microgravity exposure. The second aim will assess the suitability of a proposed set of sensorimotor assessment tasks, or measures that would be feasible within the limited time, resources, and space of a lunar/Martian lander. The assessment tasks should aid in progressive adaptation to the novel gravitational environment, provide opportunities to develop strategies to recover from off-nominal body positions, and mimic operational tasks such that crew can self-assess their potential ability to complete their missions. We will obtain performance of the sensorimotor assessment tasks at varying levels of the SDA magnitude to map sensorimotor ability to the probability of completion of operational performance measures. Our third aim involves up to 90 minutes of prolonged +3GX centrifugation to mimic the vestibular alterations observed after gravity transitions. Immediately after egress from the centrifuge, subjects will perform a subset of the sensorimotor assessment tasks followed by the operational performance tasks. We will map performance in sensorimotor assessment tasks to performance in operational performance measures, similar to aim two. Our fourth aim will utilize the NASA Johnson Space Center (JSC) Active Response Gravity Offload System (ARGOS) to characterize the effects of a reduced gravity load on balance-related exploration and operational measures. The fifth and final aim will build on the lunar landing simulations that will be developed for the Manual Crew Override study (PI: Wood) and the human lander systems training simulations to support the Flight Operations Directorate for Artemis mission training. Similar to Aim 1a, this aim anticipates working with experienced crewmembers to satisfy some of the training for generic lunar landing tasks. The study will involve two pre-test familiarization sessions focused on developing proficiency to perform the landing task through multiple landings with both the full simulator and the Just In Time (JIT) trainer platform. Following recommended guidelines for Artemis sustaining mission, crewmembers will then wait 75 days before performing the same lunar landing following a g-state analog (sustained centrifugation). The outcome of this aim is to characterize the relationship between the performance on the JIT platform with post-centrifuge lunar landing performance to inform what performance threshold is required to mitigate risk associated with vertigo and disorientation with manual crew override during landings. The main deliverables from this project will be recommended sensorimotor assessments for extravehicular activity (EVA) operations that provide a quantitative index of readiness to perform key operational tasks and validation of lunar landing simulations for training manual crew override during landings. NOTE: Per the Principal Investigator (PI): During the first project year, Aim 1c was removed from the project (Ed., 4/18/23). Aim 1c was to examine a large number of proposed sensorimotor assessment tasks in an effort to reduce the set of tasks to those that are the most sensitive to changes in sensorimotor disruption via the SDA.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This project is in direct response to the baselined Human Research Program (HRP) Path to Risk Reduction milestone of providing updates to the NASA Fitness For Duty Standards. The new standards should be tied to fitness for duty for exploration tasks and provide a quantitative index of readiness to perform key exploration tasks. This research effort will leverage expertise based upon HRP-funded flight research investigations including Functional Tasks Test, Field Test, Standard Measures, and Manual Control, as well as MedB computerized dynamic posturography. This project will leverage critical mission tasks previously established by Ryder et al. ("A novel approach for establishing fitness standards for occupational task performance." Eur J Appl Physiol, 2019) for standards related to the risk of reduced muscle mass, strength, and endurance. This project must also leverage experience with vestibular spaceflight analogs (e.g., Galvanic vestibular stimulation, sustained 3Gx centrifugation) to characterize how the deconditioned state following G-transitions (e.g., postural instability, motion sickness, head movement restrictions) map to functional performance.
Research Impact/Earth Benefits:	This research addresses the Risk of Altered Vestibular/Sensorimotor Function Impacting Critical Mission Tasks (Sensorimotor Risk) and associated research gaps from NASA's Human Research Roadmap. The development of sensorimotor assessments for EVA and manual control addresses each gap associated with the characterization of the effects of spaceflight on the sensorimotor and neuro-vestibular systems, including postural control and locomotion (Gap SM-101), manual control (Gap SM-102), spatial orientation and motion sickness (Gap SM-103), and other brain functions such as sleep, cognition, attention (Gap SM-104). This project is in direct response to the baselined Human Research Program (HRP) Path to Risk Reduction milestone of defining sensorimotor assessments that can provide a quantitative index of readiness to perform key exploration tasks. The Flight Operations Directorate has identified the need for better sensorimotor assessment tools that would protect against hazards of performing operational tasks too soon following gravitational transitions. The project outcomes will address this recommendation by providing a validated set of sensorimotor assessment tasks that will 1) allow for self-assessment when communication with ground support may be limited or delayed and 2) provide progressive adaptation to the novel gravitational environment. For Earth-based benefits: The sensorimotor community has a need for simple assessment tasks that could be used clinically, through telemedicine, or by the patient alone. These tasks could be used to help in self-determination of the patient's sensorimotor ability to perform daily activities and/or guide recovery of vestibular-impaired patients.

Task Progress:

During the second project year, Aim 1a results were submitted to Frontiers in Physiology, Aim 1b results were presented at the Human Research Program Investigators' Workshop (IWS) 2024, Aim 2 Phase 1 was completed, and Aim 2 Phase 2 and Aim 3 are underway.
Aim 2 This aim seeks to map performance in a proposed set of sensorimotor assessment tasks with performance in operational analog tasks. Validation of the proposed set of sensorimotor assessment tasks will ultimately down select the task set to those that meet the following criteria: 1) have a strong correlation between performance in the sensorimotor assessment tasks with operational analog tasks, and 2) elicit distinct performance groupings across varying levels of sensorimotor disorientation. Aim 2 is the first validation study in the overall project. The deliverable from this aim is to 1) provide an initial performance prediction model between the sensorimotor assessment tasks and operational analog tasks, and 2) define performance thresholds across a range of sensorimotor disorientation. The model generated from Aim 2 will be further refined utilizing data from Aims 3 and 4. Therefore, it is important to ensure the model is generalizable to new data.

In discussions with the Flight Operations Directorate, feedback from a Technical Interchange Meeting sponsored by the Human Research Program (HRP) Health Human Countermeasures Element, and lessons learned from Apollo Missions, the proposed set of sensorimotor assessment tasks was developed to include the following parameters: 1) mimic body maneuvers such as reaching, bending over, etc., such that crew can self-assess their potential ability to complete operational tasks; 2) provide opportunities to develop strategies to recover from off-nominal body positions; and 3) aid in progressive adaptation to the novel gravitational environment. Operational logistic constraints included: 1) tasks should be completable within a limited space and within the various Human Lander System (HLS) configurations, and 2) complete tasks with minimal to no hardware mass utilizing the available HLS configurations. The sensorimotor assessment tasks were defined and completed in the following order, starting from the least vestibularly provocative movements: postural challenge task, step test (Eymir et al., 2022; Feld et al., 2022; Hong et al., 2012; Mercer et al., 2009), fall recovery task from prone and supine, four square step test (Dite and Temple, 2002; Whitney et al., 2007), an obstacle turn task required 180- and 90-degree turns, kneel-and-turn task, and an augmented reality full body eye-hand coordination task simulating an EVA geological sampling task.

Aim 2 utilized high-fidelity operational analog tasks that were developed to simulate top-hatch capsule egress and early exploration EVA (Egress Fitness, PI: Norcross).

A total of 21 subjects completed Aim 2 Phase 1. Subjects attended two sessions separated by a minimum of 1 day. The first session provided familiarization to the sensorimotor assessment tasks and operational analog tasks and then baseline data were collected. The second session completed the assessment and operational tasks under two levels of sensorimotor disorientation: 1) high-level replicating crewmember immediate postflight performance (R+0), and 2) low-level replicating R+1 when crewmembers are within the recovery phase yet still have slight sensorimotor disorientation. To achieve these levels of disorientation, the validated Sensorimotor Disorientation Analog (SDA) was used (Moudy et al., 2023; Moudy et al., 2024). The SDA is comprised of galvanic vestibular stimulation simulating vestibular disruption (high: peaks reach 3mA, low: peak reach 2mA) and a weighted suit at the chest and distally at the ankles and wrists simulating proprioceptive deconditioning and subjective heaviness (high: 30% added bodyweight, low: 15% bodyweight). A cross-balanced approach was used where half of the subjects experienced the high level first followed by the low level and vice versa for the other half of subjects.

An interim analysis was conducted on the initial sample population of 21 subjects to assess the R-squared value and determine whether the planned sample size was appropriate to achieve a high precision efficiency (PE). PE refers to the ability of a sample-derived prediction equation to extrapolate to future samples (i.e., generalizability), with higher numbers indicating better extrapolation. Mixed linear models were conducted for two operational performance measures individually, which yielded R-squared values of 0.47 (operational task: time to complete capsule egress) and 0.83 (operational task: number of bag passes during EVA traversal task). These values indicated a sufficient level to achieve a high PE. The sample size calculation based on a number of predictors, effect size, and PE indicated a required sample size of 37 to achieve a PE of 0.8, assuming 2 predictors in the model (SDA level and one assessment task) and an R-squared value of 0.47.

The interim analysis secondarily assessed if distinct performance groupings existed across the SDA levels. A repeated measures ANOVA was conducted. All ANOVA tests, except for the time to complete the task board, were significant, indicating that at least two of the groups had different means. Paired t-tests were performed to compare any two groups. These data were also used to conduct a secondary sample size estimation. For the operational capsule egress task, assuming a mean difference of 5 and a standard deviation of 10 to detect a one-side paired difference between groups, a sample size of 38 would be needed to achieve 80% power at an alpha level of 0.0167 (0.05/3). A smaller sample size would be needed if we calculate it based on summary statistics of other variables.

Based on the interim analysis recommendations, an additional 16 subjects (a total of 37 needed minus initial 21) were required to achieve a precision efficiency (PE) of 0.8. Data collection is ongoing for 20 additional subjects (total N: 41 subjects) to allow for any dropouts (conservative 25% attrition rate) to meet sample size estimations required to develop a generalizable prediction model. This Aim 2 Phase 2 is expected to be completed in May 2024. An additional analysis will be completed again to confirm whether we have met the required sample size given the additional datasets.

A second finding of this analysis was the inability of the simulated EVA task board portion to measure relevant sensorimotor changes in performance. No significant differences were found between SDA levels for this task and correlation values to the sensorimotor assessment tasks were weak (Pearson r = 0.35). This was not unexpected as the task has an inherent stabilizing feature where subjects hold onto the umbilicals and task board to complete the task. A light touch has been shown previously to provide enough stabilizing support to compensate for vestibular disruptions (Baldan et al., 2014). As such, and in preparation for future aims where the transportability of operational analog tasks is critical, data collection for the additional 20 subjects is being conducted using updated operational analog tasks. The operational analog tasks include the same simulated capsule egress, and an alternative simulated EVA traversal that replicates the key physiological requirements as the original EVA traversal task.

Aim 3 Similar to Aim 2, Aim 3 is focused on mapping the relationship between performance in the sensorimotor assessment tasks to performance in the operational analog tasks under varying levels of sensorimotor disorientation. Aim 3 will induce sensorimotor disorientation through sustained hypergravity centrifugation which drives vestibular adaptive changes similar to G-transitional effects following spaceflight (Albery and Martin, 1996; Bles et al., 1989; Groen et al., 2008; Nooij et al., 2007). The centrifuge exposure parameters are based largely on the protocol established by Groen and TNO colleagues in Soesterberg, The Netherlands. Based on their work, the optimal exposure is +3Gx for 60 min

(Bles et al., 1997; Nooij and Bos, 2007). Data collection will occur at the KBR Aerospace Environment Protection Lab (AEPL) located in San Antonio, TX. The KBR AEPL centrifuge has a free-swinging gondola at the end of the long arm, resulting in a constant alignment of the gravito-inertial force with the gondola. The subject will be oriented in a supine position in the gondola (feet pointing in the direction of motion) so the resultant force vector is in the naso-occipital direction to minimize orthostatic stress. Although the centrifuge is capable of fast G-onset for performance jet training, our profile will utilize a moderate 0.1 G/s acceleration and deceleration rate.

Subjects will attend two sessions separate by a minimum of one day. The first session will be to familiarize the subjects with the tasks and complete a short 5-minute centrifuge exposure to minimize biases toward prior exposure. The centrifuge profile used in this study is unique to the AEPL facility and subject pool, therefore, we do not expect any data skew due to prior exposure. The second session will be the test session to include pre-exposure baseline data collection, followed by a sustained 60-minute exposure, and two post-centrifuge tests separated by 45 minutes. The test session was developed such that the post-centrifuge tests mimic the two postflight time points and SDA levels (R+0/high, R+1/low) from Aims 1 and 2.

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Bibliography Type:	Description: (Last Updated: 04/01/2026)
Abstracts for Journals and Proceedings	Moudy SC, Peters BT, Clark TK, Schubert MC, De Dios Y, Bollinger A, Wood SJ. "Validation of a sensorimotor disorientation ground analog." 2024 NASA Human Research Program Investigators' Workshop. Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop. Galveston, Texas, February 13-16, 2024. , Feb-2024
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